



Startup Zone

- 1 Sustainable Water Treatment With CDI
- 2 Holograms For Driver Navigation
- 3 Drive To The Sky With SkyDrive
- 4 An AI Companion For Kids
- 5 An Automated Burger-Making Robot

1 SUSTAINABLE WATER TREATMENT WITH CDI



Water supports life of all organisms on Earth, and it is essential to drink safe water for healthy living. Drinking contaminated water causes a host of diseases such as typhoid, dysentery and cholera, which can be life-threatening. InnoDI (Innovative De-ionisation) Water Technologies, an IIT-Madras incubated startup, was founded as a joint venture between Milano-based Idropan Dell'Orto Depuratori and Bengaluru-based AquaSphere Greentech Solutions in 2016. Their aim is to globally deploy Capacitive Deionisation (CDI) technology for economical and sustainable water treatment by decreasing water and energy wastage (with DC power) when compared to traditional RO systems.

The employed water treatment technology, CDI, removes salinity from ground or surface water with-

out using any chemicals (resins) or membrane filters. The manufacturing plant near Bengaluru has the capacity to produce CDI cells at high volumes based on customer requirements. The cells have numerous carbon-based electrodes through which water is sent at low pressure, and a potential difference is applied between the electrodes (forming a capacitor) to dissociate the ions present in water.

At a voltage difference of 1.6V, direct current results in an electrostatic field that pulls the ions towards oppositely-charged electrodes. Low voltage prevents electrolysis and instead results in separation of salts and metals from water. Once electrodes get saturated with ions, their polarity automatically gets reversed. So, ions fall off electrodes, and water with dissolved ions gets drained out of the system.

Then, polarity gets reversed again, and the cycle continues. For avoiding precipitation in cells and thus limiting their life, the company's patented maintenance solution is automatically injected from time to time depending on inlet water conditions.

The USP is that with incorporation of smart electronics and plug flow concept, cells get utilised to deliver maximum efficiency. Besides, unlike RO, control of removal rates and selective removal of only harmful toxins from the water is possible with CDI.

Applications include digital kiosks, dispensers, attachments for consumer electronics like washing machines, point-of-entry-based treatment for residential areas, point-of-use-based systems for homes and requirements for special places like in clinics. The Internet of Things (IoT)-enabled system and platform make it possible to remotely track performance based on such parameters as water quality, amount of water produced and so on.

Regarding pricing, Vijay Sampath, managing director, InnoDI Water, says, "Our product pricing is based on capacities and starts at ₹ 25,000 for a home unit, and goes up to ₹ 800,000 for a community drinking water application."

Despite having received recognition from Ministry of Drinking Water and Sanitation, government of India, when it comes to the support extended, Sampath says, "It is quite disappointing that the government that is supposed to adopt new technologies is still investing a lot of money on old ones."

Talking about future plans, Sampath says, "We have over 300 installations of our technology and these